

MTH204 (Section A)
Graded Worksheet–5

Name:

Roll No.:

Important Notice: The use of electronic gadgets is strictly prohibited during the tutorial. This includes mobile phones, tablets, or any device containing class notes, but anything handwritten is completely allowed.

Q–1) Solve the I.V.P:

$$(D^3 - 4D)y = 10 \cos x + 5 \sin x,$$
$$y(0) = 3, \quad y'(0) = -2, \quad y''(0) = -1.$$

Q–2) Solve the ODE:

$$x^3 y''' + 2x^2 y'' - xy' + y = \frac{1}{x^2}.$$

Q–3) Assume that all the zeros of the polynomial

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

have negative real parts. If $u(t)$ is any solution to the ODE

$$a_n \frac{d^n u}{dt^n} + a_{n-1} \frac{d^{n-1} u}{dt^{n-1}} + \cdots + a_1 \frac{du}{dt} + a_0 u = 0,$$

then

$$\lim_{t \rightarrow \infty} u(t) = \underline{\hspace{2cm}}.$$

Q–4) The boundary value problem

$$x^2 y'' - 2xy' + 2y = 0, \quad 1 \leq x \leq 2,$$

$$y(1) - y'(1) = 1,$$

$$y(2) - k y'(2) = 4,$$

has infinitely many distinct solutions when

$$k = \underline{\hspace{2cm}}.$$

Q-5) Prey-Predator Model.

Two species interact in an ecosystem: a prey population $x(t)$ and a predator population $y(t)$.

Assume:

1. In the absence of predators, the prey grows at a rate proportional to its population.
2. In the absence of prey, the predator population decays at a rate proportional to its population.
3. The interaction between predator and prey is proportional to the product of their populations.

Using these assumptions, formulate the system of differential equations describing the dynamics of the populations.

Solution:

Solution:

Solution: